

Teacher Information, Course Description and
Lecture-wise Plan
of
Operating Systems

under

Pilot Project:
Uniform Teaching, Assessment and Quality
Assurance Across All The Campuses

by

Faculty of Information Science and Technology
COMSATS University Islamabad

A: Teacher Information

Course Name: Operating System	Credit Hours: 2
Course Code:(CSC323)	Pre-requisite:None
Teacher Name: Khalid Awan	E-mail: Khalid.awan@cuiatk.edu.pk
Program: BS (Computer Science)	Semester: Fall 2026
Department: Computer Science	Campus: CUI

B: Course Description**Topics Covered:**

This course introduces the services and functions performed by operating system for smooth and accurate system operations. Topics include: Operating Systems Overview; Device Organization & System Operations; Operating Systems Principles; Process Management; Process Synchronization; Deadlocks; Multiprocessor Issues; Memory Management; Storage Management; Security & Protection; case studies of desktop and mobile systems.

Textbook:

1. Operating System Concepts, Silberschatz & Galvin, Addison-Wesley, 2021.

Reference Book:

2. Operating Systems: Internals and Design Principles, Stallings, W., Pearson, 2017.
3. Modern Operating Systems, Tanenbaum, A. S., Prentice Hall, 2014.

C: Lecture-wise Activities

Week #	Lecture #	Topic (Chapter/Topics Covered)	Evaluation	Remarks
1	1	Overview of Operating System: - Operating system - User View - System View - Computer System Organization - Interrupts and its implementation - I/O Structure - Computer System Architecture - Single processor System - Multiprocessor system - Clustered System - Operating system operations - Multiprogramming & Multitasking - Dual mode and Multimode		Silberschatz: Ch1
	2	Device Organization & System Operations: - Computing Environments - Traditional - Mobile - Client-Server - Peer to peer - Cloud - Interrupts - Interrupt-driven operation; - Types of Interrupt		Silberschatz: Ch1
2	3 (Date)	Operating System Structure & Services: - OS services - User and OS Interfaces - Command Interface - Graphical User Interface - Mobile - Touch Screen - Choice of Interface - System calls		Silberschatz: Ch2
	4 (Date)	OS Structure & Design Issues: - OS Design and Implementation - OS Structure - Monolithic - Layered - Microkernel	Assignment#1	Silberschatz: Ch2

		○ Modular ○ Hybrid Syste		
3	5 (Date)	• OS security and Protection mechanism	Quiz # 1	Silberschatz: Ch16&17
	6 (Date)	Process Management: • Process concept ○ Process State ○ PCB • Process Scheduling ○ Scheduling Queues ○ Process Scheduling ○ Context Switch • Process Operations ○ Process Creation (Fork, Exec & Wait System-Calls) ○ Process Termination (Exit & abort system calls, Cascading, Termination, and Zombie & Orphan Processes)		Silberschatz: Ch3
4	7 (Date)	Inter-Process Communication: • Inter-process communication (IPC) ○ IPC in shared memory ○ IPC in message Passing • Sockets(Remote Procedure Call)	Exercises as given in the books	Silberschatz: Ch3
	8	Threads and Concurrency: • Threads(Overview) • Processes vs. Threads, • Multiple Simultaneous Computations • Alhamad's Law		Silberschatz: Ch4
5	9 (Date)	Threads and Concurrency: • Multicore Programming & its Challenges • Multi Treaded Models • Multi Treaded Libraries	Assignment#2	Silberschatz: Ch4
	10 (Date)	CPU Scheduling: • CPU burst/I/O burst • CPU scheduler • Preemptive vs non-Preemptive • Dispatcher • Scheduling Criteria • FCFS		Silberschatz: Ch5
6	11	Scheduling Algorithms: • SJF		Silberschatz: Ch5

	(Date)	• <i>SRTF</i> • <i>Round Robin</i> • <i>Time quantum and context switch</i> • <i>Priority scheduling</i> • <i>Priority scheduling vs Round Robin</i>		
	12 (Date)	Scheduling Algorithms: • <i>Multilevel Queue</i> • <i>Multilevel Feedback Queue</i> • <i>Thread Scheduling, Pthread Scheduling</i> • <i>Discussion on scheduling algorithms used in windows and linux</i>	Quiz # 2	<i>Silberschatz: Ch5</i>
7	13 (Date)	Synchronization: • <i>Race Condition</i> • <i>Critical section problem</i> • <i>Interrupt based solution vs software solution</i> • <i>Petersons solution</i> • <i>Petersons solution revisited with memory barrier</i>		<i>Silberschatz: Ch6</i>
	14 (Date)	Process Synchronization: • <i>Hardware instructions</i> • <i>Test & set, compare & swap</i> • <i>Atomic Variables</i> • <i>Mutex Locks</i> • <i>Semaphores</i> • <i>Semaphore implementation</i> • <i>Monitors</i>		<i>Silberschatz: Ch6</i>
8	15 (Date)	Classical Synchronization Problems: • <i>Bounded Buffer Problem</i> • <i>Readers Writers Problem</i>		<i>Silberschatz: Ch7</i>
	16	Classical Synchronization Problems: • <i>Dining Philosophers problem</i> • <i>Monitors solution to Dining Philosophers</i> • <i>Synchronization in windows and linux.</i>		<i>Silberschatz: Ch7</i>
9	17 (Date)	<i>Midterm Examination</i>	Midterm (25%)	

	18 (Date)	<i>Midterm Examination</i>		
10	19 (Date)	Deadlocks: • <i>Deadlock: Definition and real-world/system examples</i> • <i>Necessary Conditions for Deadlock: Mutual Exclusion, Hold-and-Wait, No Preemption, Circular Wait</i> • <i>Resource-Allocation Graph (RAG): Notation, request edges, assignment edges</i> • <i>Interpreting Cycles in a RAG (cycle vs. guaranteed deadlock, single vs. multiple instances)</i>	Assignment#3	<i>Silberschatz: Ch8</i>
	20 (Date)	Deadlock Avoidance: • <i>Deadlock Avoidance: Concept of Safe State and Safe Sequence</i> • <i>Resource-Allocation Graph Algorithm Banker's Algorithm (multiple instances per resource type)</i>	<i>Exercises as given in the books</i>	<i>Silberschatz: Ch8</i>
11	21 (Date)	Deadlock Detection, Recovery & Case Study: • <i>Deadlock Detection: Single instance per resource type : Wait-for Graph and cycle detection</i> • <i>Deadlock Detection: Multiple instances per resource type: Detection Algorithm (Available, Allocation, Request)</i> • <i>Detection-Algorithm Usage</i> • <i>Recovery from Deadlock</i> • <i>Case Study: Process synchronization & deadlock handling in Windows OS</i>		<i>Silberschatz: Ch8&21</i>
	22 (Date)	Memory Management: • <i>Background: Physical Memory and Memory Management Hardware (</i> • <i>Base and Limit Registers, Memory Protection</i> • <i>Address Binding: Compile-Time, Load-Time, and Execution-Time binding</i> • <i>Logical vs. Physical Address Space; the MMU and the Relocation</i>	Quiz # 3	<i>Silberschatz: Ch9</i>

		<i>Register</i> • <i>Dynamic Loading: loading routines only when called</i> • <i>Dynamic Linking and Shared Libraries</i> • <i>Windows Example: DLL (Dynamic Link Library) loading and shared-library resolution at run time</i>		
12	23 (Date)	Contiguous Memory Allocation & Segmentation: • <i>Contiguous Memory Allocation: Fixed Partitioning (MFT)</i> • <i>Contiguous Memory Allocation: Dynamic Partitioning (MVT)</i> • <i>Allocation Strategies: First-Fit, Best-Fit, and Worst-Fit</i> • <i>Fragmentation: Internal vs. External, and Compaction</i> • <i>Non-Contiguous Allocation: Segmentation: basic method and user's view of memory</i> • <i>Segmentation Hardware: Segment Table, Base and Limit per segment, protection bits</i>		<i>Silberschatz: Ch9</i>
	24 (Date)	Paging: Basic Method & Hardware Support: • <i>Paging: Basic Method & Hardware Support</i> • <i>Paging: Basic Method: Page Table, Frame Table, address translation</i> • <i>Hardware Support: Translation Look-aside Buffer (TLB) and Effective Access Time</i> • <i>Protection in Paging: Valid-Invalid Bit, Read/Write/Execute permission bits</i> • <i>Advanced: Hashed Page Tables and Inverted Page Tables</i>		<i>Silberschatz: Ch9</i>
13	25 (Date)	Virtual Memory & Demand Paging: • <i>Virtual Memory: Background and Benefits over pure physical-memory schemes</i> • <i>Demand Paging: Basic Concept and Pure Demand Paging</i> • <i>Page-Fault Handling: step-by-step sequence of events</i>		<i>Silberschatz: Ch10</i>

		• <i>Performance of Demand Paging: computing Effective Access Time (EAT)</i> • <i>Copy-on-Write: sharing pages between parent and child processes</i>		
	26 (Date)	Page Replacement Algorithms: • <i>Overview</i> • <i>FIFO</i> • <i>Optimal</i> • <i>LRU</i> • <i>SecondChance Algorithm</i> • <i>MFU & LFU</i> • <i>Thrashing.</i>	<i>Exercises / page-reference strings</i>	<i>Silberschatz: Ch10</i>
14	27 (Date)	Case Studies: • <i>Memory Management in Windows and on ARM architecture</i>	<i>Assignment#4</i>	<i>Silberschatz: Ch9&21</i>
	28 (Date)	Mass Storage Management: • <i>Hard disks</i> • <i>Volatile and non-volatile memory</i> • <i>Disk scheduling</i> • <i>NVM scheduling</i>	<i>Quiz # 4</i>	<i>Silberschatz: Ch11</i>
15	29 (Date)	Storage Device Management: • <i>Formatting</i> • <i>Boot Block</i> • <i>Bad Blocks; and Swap-Space Management.</i>		<i>Silberschatz: Ch11</i>
	30 (Date)	File System: • <i>File concept</i> • <i>File attributes and operations</i> • <i>Access methods</i> • <i>Directory structure</i> • <i>File-System Structure</i> • <i>File System Operations</i>		<i>Silberschatz: Ch13</i>
16	31 (Date)	File System: • <i>Directory Implementation</i> • <i>File Protection; and Memory-Mapped Files.</i>		<i>Silberschatz: Ch13</i>
	32 (Date)	Case studies: • <i>Windows and Apple file system</i>	<i>Final Review / Examination Preparation</i>	<i>Silberschatz: Ch14&21</i>
17	(Date)		Final (50%)	

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